Protocol for amino acid refinement through computational evolution

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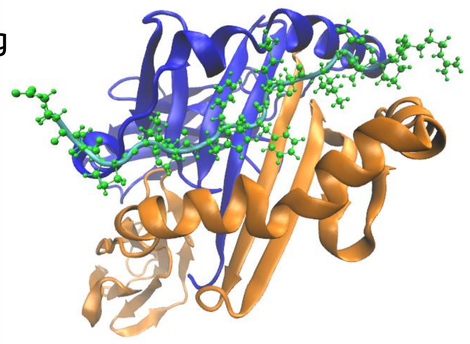
(Group's website)

Motivation

- Design peptides (and proteins) with better affinities towards a particular target, including proteins. The use of structural information allow the combination of *de novo* and template-based approaches [1]

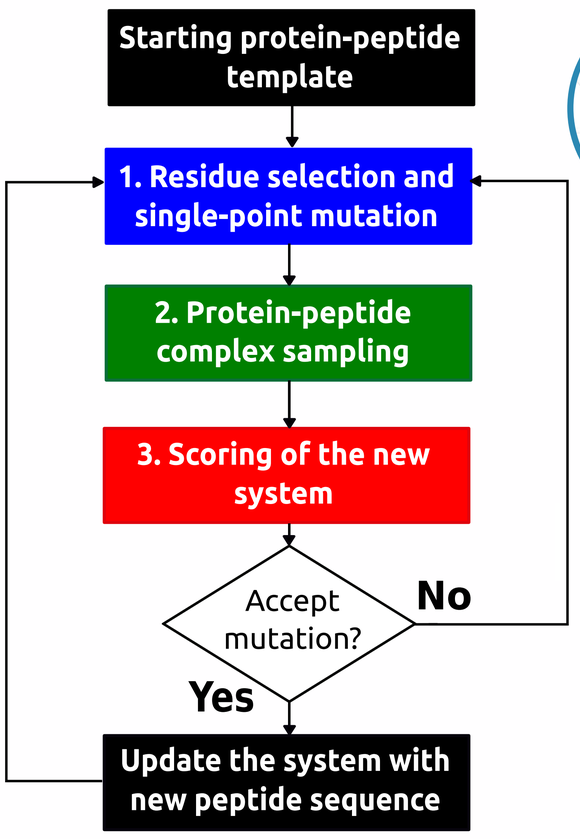
Goal: Evolve bound peptides based on single-point mutations, sampling of the conformations and scoring strategies



Protocol

Template peptide

AAPAAAPP



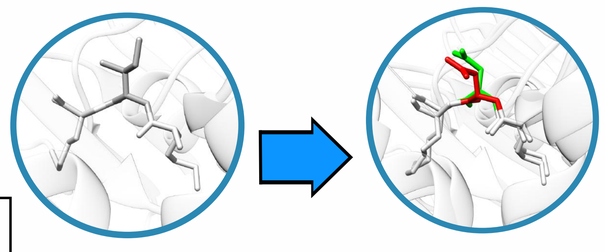
Predicted peptide

YIWMFHDM

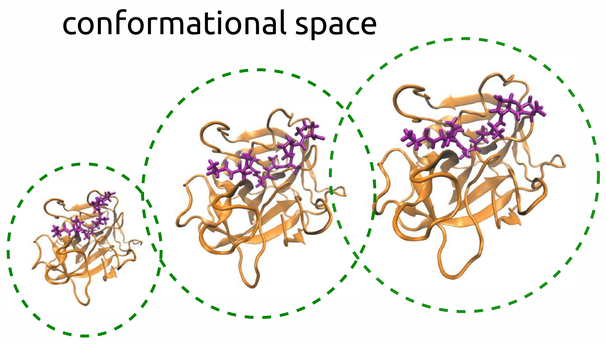
- The acceptance criterion is a consensus approach with multiple scoring functions

Six scoring functions: BACH [2], ZRANK [3], Firedock [4], Pisa [5], BMF-BLUUES [6] and IRAD [7] **are included within the code**

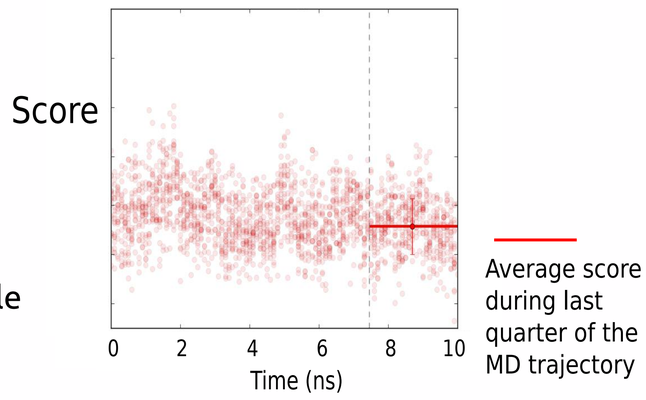
- Prediction of rotamers frequently sampled by MD simulations



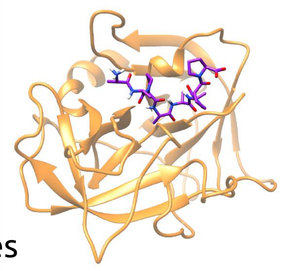
- Efficient MD simulations after each mutation to explore the conformational space



- Scoring functions sensible to single-point mutations

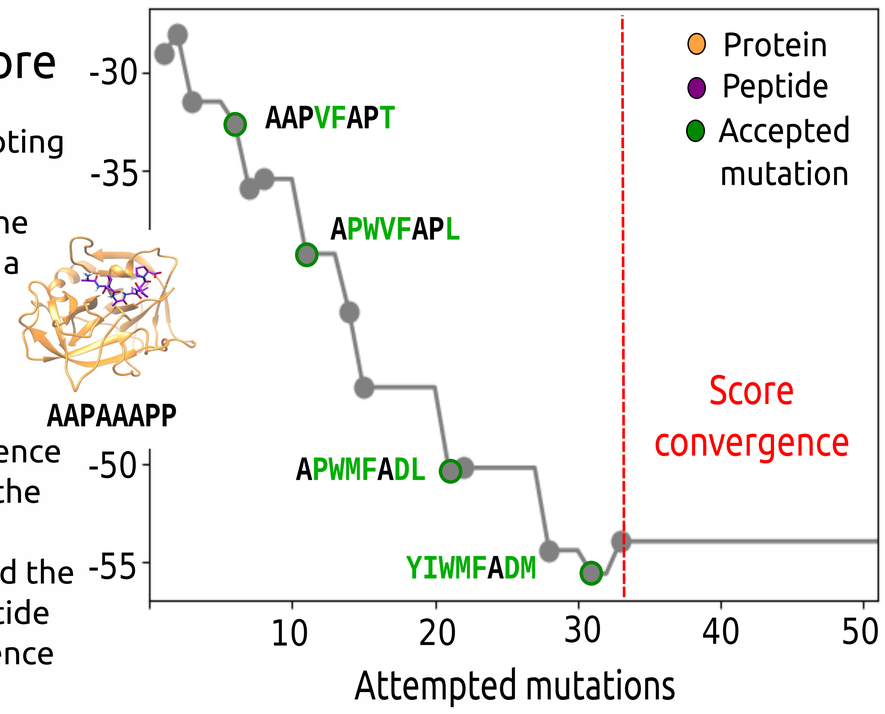


- The protocol requires of a protein-peptide complex that can be crystallized or modelled based on docking methodologies



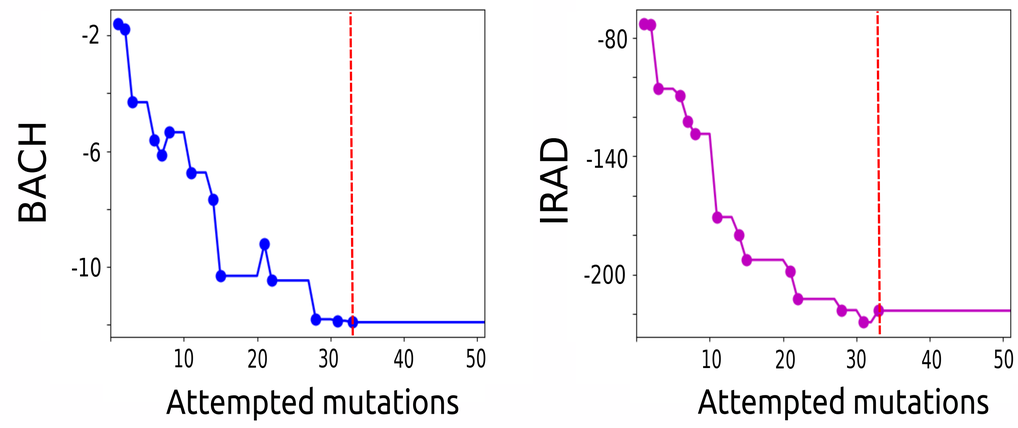
Validation

- After attempting a number of mutations, the scores reach a plateau



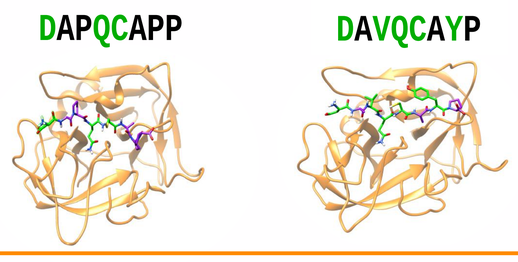
- The convergence depends on the number of attempts, and the starting peptide bound sequence

- Based on the scoring consensus criterion, multiple scoring functions are implemented to verify **if their values are minimizing while accepting a set of the mutations and new peptide sequences**



The protocol requires as input:

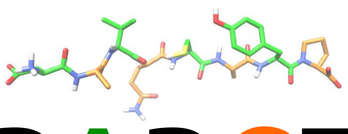
- PDB files and chain identifiers
- Residues modified in the peptide
- Number of mutations
- List of scoring functions
- Consensus threshold
- Files names and paths
- Modes to run the protocol



All the outputs regarding the trajectories, conformations and scores are stored for further analysis

Project code

https://github.com/PARCE-project/PARCE-1



PARCE

Protocol for **A**mino acid **R**efinement through **C**omputational **E**volution

Dependencies/ Third-party tools:

python, #!/bin/bash, biopython, docker, GROMACS FAST. FLEXIBLE. FREE., SCWRL4

Conclusions

- Our protocol, called PARCE, can accelerate the discovery of novel peptides as potential therapeutics, biomarkers or vaccine sub-units. The code is flexible, allowing adding protocols to modify different types of molecular targets, such as small molecules
- Overall, the protocol contributes to the engineering of peptides and proteins for a broad spectrum of applications

References

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